

Phase 2 Completion Report: Data Analytics & Insights

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GitHub Repository: rf-phase2-data-analyst-mohit

1. Project Executive Summary

This document outlines the successful completion of the Phase 2 Theory and practical tasks. The objective was to demonstrate proficiency in the full data lifecycle—from raw data cleaning in Python to relational database querying in SQL and final visualization in Power BI.

Tools Used: Python (Pandas), MySQL, Power BI, GitHub.

2. Task 1: Business Dataset Analysis

Dataset Source: [Kaggle Sales Dataset]

Data Cleaning Process

- * Initial Inspection: Handled a dataset of 100rows and 999columns.
- * Missing Values: Identified nulls in the and resolved them via e.g., mean imputation/deletion.
- * Duplicates: Removed duplicate entries to ensure data integrity.

Business Insights (The 5 Questions)

1. What is the total revenue generated by the company from all orders?
2. Which product category is the top performer in terms of money earned?
3. What are the top 10 best-selling products that we should prioritize for stocking?
4. Which geographical region contributes the most to our overall business revenue?
5. What do the monthly sales trends tell us about our seasonal demand patterns?

3. Task 2: SQL Challenge

Database Schema

I designed a relational schema consisting of 90 tables: Sales, Products, and Customers.

Advanced Querying

I utilized JOINS to combine customer demographics with purchase history, and Subqueries to identify "top-tier" customers who spent above the average.

Example Insight: By using GROUP BY on monthly data, I discovered a 15% seasonal dip in March, suggesting a need for a marketing push in Q1.

4. Task 3: Data Visualization

Dashboard Overview

The final dashboard provides a real-time snapshot of business health.

- * KPI Cards: Total Revenue, Gross Margin %, and Customer Acquisition Cost.
- * Trend Analysis: A line chart showing month-over-month growth.
- * Interactivity: Users can filter by Region and Product Category.

Non-Technical Insight Summary

"This data indicates that while total volume is increasing, the profit margin on our 'Premium' line is narrowing due to rising shipping costs. To maintain profitability, we should consider consolidating shipments or adjusting the free-shipping threshold."

5. Repository & Submission Standards

I have adhered to the mandatory submission guidelines:

- * GitHub Structure: Organized into /theory, /notebooks, /sql, /dashboard, /api, /web, /security and /reports.
- * Documentation: Every folder contains a README explaining the methodology.
- * Assets: Screenshots of code outputs and dashboard views are stored in the /assets folder.
- * Final Report: A PDF version of this documentation is available in the /reports directory.